Confidence & Sentiment Analysis – Zecpath AI

Objective

To evaluate communication quality and behavioral indicators during AI-driven HR screening interviews.

The module analyzes:

- Candidate confidence
- Sentiment
- Hesitation patterns
- Communication quality
- Uncertainty signals
- Contradictions in responses

Deliverables

1. Confidence Analysis Logic

File

```
screening_ai/confidence_engine.py
```

Features

- Hesitation detection
- Response length analysis
- Speaking pace estimation
- Final confidence scoring

Confidence Formula

```
Confidence =
(Length × 0.4) +
(Pace × 0.4) +
(1 - Hesitation × 0.2)
```

Output Example

```
{
    "confidence_score": 0.82,
    "signals": {
```

```
        "hesitation": 0.1,
        "length_score": 1.0,
        "pace_score": 0.8
    }
}
```

---

2. Sentiment Scoring Module

File

```
screening_ai/sentiment_engine.py
```

Features

- Positive sentiment detection
- Negative sentiment detection
- Neutral sentiment handling

Sentiment Categories

Category	Meaning
Positive	Confident and optimistic response
Neutral	Balanced response
Negative	Weak or uncertain response

Output Example

```
{
    "sentiment": "Positive",
    "sentiment_score": 0.8
}
```

---

3. Behavioral Indicators Report

File

```
screening_ai/behavior_report.py
```

Features

- Combines confidence + sentiment analysis
- Detects uncertainty
- Detects contradictions
- Generates communication strength level

Communication Strength Levels

Score	Level
≥ 0.75	Strong
0.5 – 0.74	Moderate
< 0.5	Weak

Output Example

```
{
    "confidence": {
        "confidence_score": 0.82,
        "signals": {
            "hesitation": 0.1,
            "length_score": 1.0,
            "pace_score": 0.8
        }
    },
    "sentiment": {
        "sentiment": "Positive",
        "sentiment_score": 0.8
    },
    "behavior_flags": {
        "uncertainty": false,
        "contradiction": false
    },
    "communication_strength": "Strong"
}
```

Processing Flow

```
Answer Text
    ↓
Hesitation Detection
    ↓
Length & Pace Analysis
    ↓
Sentiment Analysis
    ↓
```

```
Uncertainty / Contradiction Detection
    ↓
Behavioral Report Generation
```

Advantages

- Improves AI-based candidate evaluation
- Adds behavioral understanding
- Captures communication quality
- Enables explainable screening decisions

Limitations

- Rule-based sentiment detection
- No real audio tone analysis
- Accent and speech style variations may affect results

Future Improvements

- ML-based sentiment models
- Emotion detection
- Real-time voice analytics
- Tone and stress analysis
- LLM-powered communication scoring

Test Script

File

```
tests/test_behavior.py
```

Test Command

```
pytest tests/test_behavior.py
```

Demo Script

File

```
demo_behavior.py
```

Run

```
python demo_behavior.py
```

Expected Output


```
{
    "confidence": {
        "confidence_score": 0.88,
        "signals": {
            "hesitation": 0.0,
            "length_score": 0.7,
            "pace_score": 1.0
        }
    },
    "sentiment": {
        "sentiment": "Positive",
        "sentiment_score": 0.6
    },
    "behavior_flags": {
        "uncertainty": false,
        "contradiction": false
    },
    "communication_strength": "Strong"
}
```